

U.S. State-Level Sales & Demographic Analysis (2022–2024)

1. Business Question

This project analyzes sales performance across U.S. states by integrating transactional sales data with demographic indicators. The objective is to evaluate how factors such as population size, median income, and urbanization influence total revenue and per-capita sales, helping identify high-performing markets and the underlying drivers of demand. The analysis aims to answer key questions such as: *Which states generate the strongest per-capita sales performance, and what demographic characteristics explain these differences?*

2. Data Overview

Primary Database: dwsql_course

Tables used:

1. orders - order details, dates, statuses, shipping states
2. order_items - quantities and unit prices
3. customers - region and state validation
4. regional_sales - monthly region-level revenue

External Data: state_population.csv → Through import-wizard and insert data to sql 00_setup

1. State_population - state, population, median_income, region, urban_pct

Data Cleaning

- Standardized state field spaces using 'HAVING LENGTH () > 2' to check for states that have more than 2 letters (No states had more than 2 letters)
- Check for NULL values in orders table using 'IS NULL' (No missing values)
- Check for pricing Anomalies in Order items by setting unit_price and qty to <=0 (None of the order items have a unit price or qty of 0 or less)
- Verify data integrity with External data (No missing data)

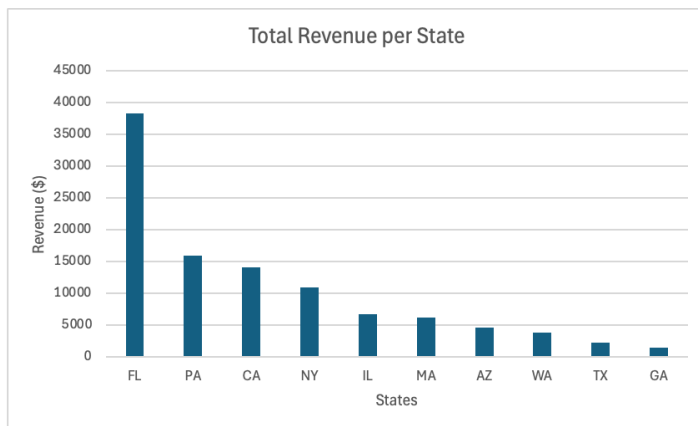
3. Key SQL Techniques Used

1. Aggregation & Joins
 - a. SUM, COUNT, AVG, GROUP BY, HAVING
 - b. Multi-table joins between orders, order_items, and population data
2. CTEs
 - a. A temporary alias was created to calculate state-level revenue and monthly sales trends, allowing these metrics to be reused across multiple CTE steps without repeating the underlying joins and aggregations.
3. Windows Functions
 - a. 'DENSE_RANK()' - per-capita ranking

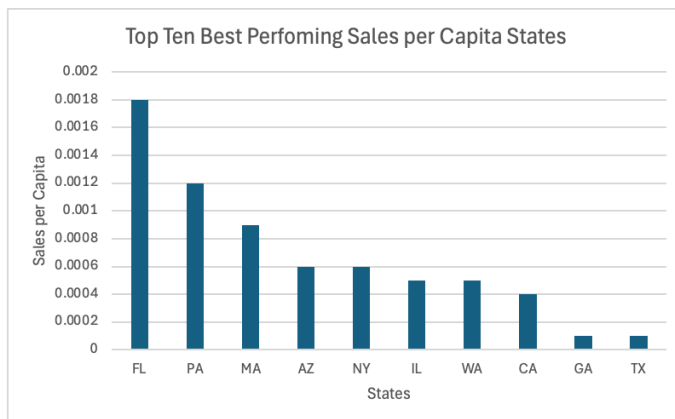
- b. 'AVG() OVER()' - 3-month rolling average
 - c. State and income segmentation analysis
- 4. Optimization
 - a. Indexing to frequently joined columns to reduce full-table scans and improve query performance, especially for joins involving customer IDs, order IDs, product IDs, and state fields.
- 5. Visualization Prep
 - a. Created a consolidated chart 'v_final_state_metrics' for chart and summary table production

4. Results

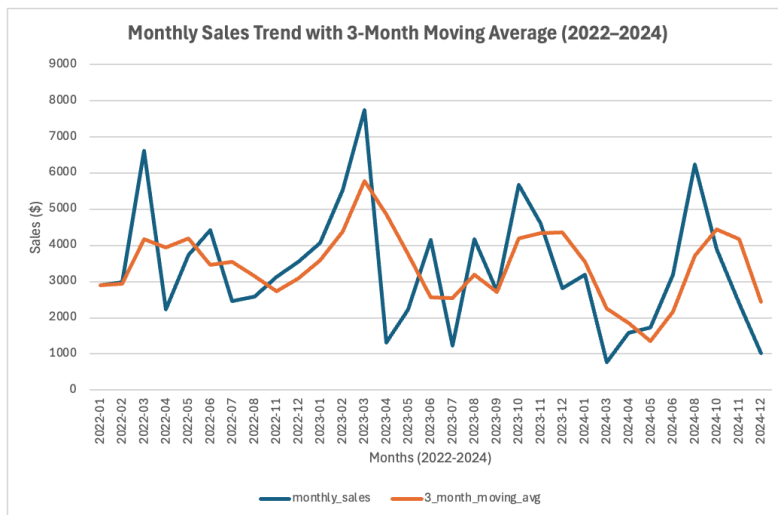
1. Total Revenue by State: A bar chart was created with the data extracted with Florida leading in revenue at \$38,327.04 also having the most orders as well at 25 orders. Georgia was the lowest performing state with a revenue of \$1,506.90 with one of the two least total orders at 2.



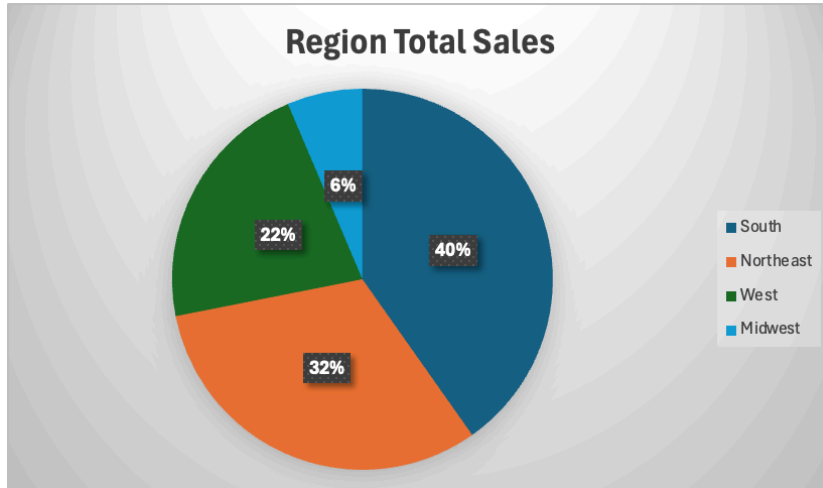
2. Rank State by Per-Capita Sales: A bar chart was created to highlight the top ten performing states. Florida was the best performing state at a per-capita of 0.0018 and Texas and Georgia the lowest performing state at a per-capita of 0.0001.



3. Revenue vs. Population Segmentation: Population was categorized into three groups, high population (>10M), medium population (5M-10M), and small population (<5M). Through analysis there are 7 states identified as high population areas, 3 states as medium population areas, and no states as low population. High population areas had a higher average revenue per state of \$12,854.06 than medium population of \$4,954.84. This indicates that states with larger populations tend to generate higher overall revenue, suggesting that market size is a key driver of total sales performance.
4. Rolling 3 Month Average: A two-series line graph is created to compare monthly income trends against monthly spending patterns, identifying gaps, alignments, and overall financial behavior over time. Monthly sales were observed to have sharp spikes and drops month-to-month with inconsistent sale patterns. The 3-month moving average reveals the underlying trend indicating growth periods in early 2023, late 2023, and mid 2024 while low periods in mid 2023, early 2024, and late 2024. These short-lived spikes suggest the presence of seasonal promotions, one-time campaigns, or irregular buying cycles that temporarily boost sales but do not sustain long-term momentum.



5. Regional Performance Comparison: A pie chart was created to compare regional sales trends. The South region had the highest total revenue of \$42,163.45 (40%) and the most amount of orders of 29 orders. While the midwest performed the lowest with a total revenue of \$6,715.46 (6%) with 6 orders. This indicates that regional demand is unevenly distributed, with the South representing a significantly larger and more active customer base, while the Midwest shows weaker engagement and smaller market potential.



6. **Impact of Income on Spending:** Medium income customers produced the highest average order value and number of orders of \$1333.68 and 71. This indicates that customers in the medium-income segment are the most engaged and exhibit the strongest purchasing power, making them a key driver of overall sales performance.

5. Limitations

The dataset used in this analysis was relatively small, containing only 30 customers and 80 orders, which limits the generalizability of the findings. As a result, the patterns observed, such as revenue differences by region or population group, reflect only the behavior of this small sample data rather than true population trends. Additionally, per-capita values were normalized for comparison purposes and do not represent actual statewide economic indicators. Since, population and sales values are based on a limited sample, per-capita values become more volatile, making certain states appear disproportionately high or low. Finally, the dataset lacks real-world complexity such as seasonality drivers, marketing impacts, and customer demographics, which restricts deeper causal analysis.

6. Next Steps

1. Expand the dataset: Incorporate more data, the larger the data the smaller the standard error, increasing reliability.
2. Add customer-level attributes: Including customer age, detailed income bracket, and household size can allow a more precise customer spending behavior.
3. Category-Specific Trends & Seasonality: Analyzing sales by product category over time to identify seasonal patterns, promotional patterns, high-demand periods, and category-driven fluctuations.
4. Predictive Modeling: Predictive models can be developed to forecast per-capita performance and regional sales growth. Help quantify how population, income, and urbanization influence future demand. This would support strategic planning, inventory decisions, and targeted regional marketing initiatives.